Week 3 | Intelligent Agents & Environment Modeling

CLO - 3: *Design basic AI-based solutions for real-world problems using appropriate techniques such as reasoning, tracking, or robot localization.*

LAB OBJECTIVE

1. To Create AI Workflow using Node automation tool
2. To implement and analyze fundamental intelligent agent architectures and environment models using Python, emphasizing computational reasoning, agent–environment interaction, and decision-making behavior.

Submission Guidelines:

1. Download the .ipynb file from Jupyter Notebook and named it as **yourRollNumber_Lab3_Agents**.
2. Create a **Word report** documenting **all the steps** you performed while creating the **n8n workflow**.
   - Include **screenshots for each step** showing both **configuration** and **successful execution**.
   - Save the document as **PDF**, and upload it to **GCR**.
3. Place screenshots clearly under their respective steps.
4. **Do NOT Crop** any screenshot: Your full screen should be visible.
5. **Blurry Screenshots** will not be considered.
6. Submit the .ipynb and .pdf Files individually (zip File will NOT be considered) to GCR Only.
7. Submission Via any other platform will NOT be Marked.
8. **Failure to follow the submission guidelines will result in a deduction of 50% of the total marks.**

PART A: N8N

TASK 1: INTELLIGENT EMAIL WORKFLOW | AI-DRIVEN INBOX ORCHESTRATION AGENT (USING N8N)

Case Study

Background

XYZ is a faculty member in the Department of Artificial Intelligence & Data Science. On a daily basis, she receives a high volume of heterogeneous email traffic, including:

- Lab-related emails from students that frequently contain missing information, unclear subject lines, or incomplete explanations
- Departmental and administrative emails concerning meetings, scheduling, and time-sensitive academic matters
- Miscellaneous emails that do not contribute to her instructional or academic responsibilities

Manual inspection, prioritization, and response formulation for such emails introduces significant cognitive overhead and adversely impacts instructional productivity. *To mitigate this inefficiency, an **automated, AI-assisted email management agent** is required.*

Objective

Design and implement an **AI-powered email orchestration workflow** using **n8n** that autonomously:

- Performs **semantic intent classification** of incoming emails
- Dynamically routes emails based on inferred intent
- Generates contextualized draft responses only when necessary
- Applies priority-based labels without unnecessary reply generation

Functional Requirements

The system must classify all incoming emails into **exactly one** of the following semantic categories:

- **Lab_Query**
- **Departmental_Meeting (High Priority)**
- **Ignored**

Technology Stack

- **Gmail Trigger Node:** Event-driven detection of new inbound emails
- **Set Node(Edit Field):** Input sanitization and feature reduction
- **OpenAI (GPT):** Natural Language Understanding (NLU) for intent classification
- **IF Node**: Conditional branching and control flow logic
- **OpenAI (GPT)**: Automated professional response synthesis
- **Gmail Draft Node**: Draft persistence without transmission
- **Gmail Label Node**: Priority-based inbox annotation

Workflow Specification

1. Event Trigger

Initiate workflow execution upon receipt of a new email in the Gmail inbox.

2. Input Preprocessing (Set Node)

Extract and retain only semantically relevant fields for downstream NLP processing:

- Email body
- Sender address
- Subject line

All extraneous metadata must be discarded to minimize prompt noise.

3. AI-Based Intent Classification (OpenAI Node: select "Message a Model")

Apply a large language model to infer the semantic intent of the email. First claim the credits, then choose ***GPT-5-MINI*** model.
Select a role: System

System Prompt (STRICT: USE AS IS)

You are an email classifier for an academic instructor named XYZ.

Classify the email into ONLY one of the following labels:

1. Lab_Query

2. Departmental_Meeting

3. Ignored

Rules:

- Lab_Query: student asking about lab work, submissions, issues, marks, deadlines

- Departmental_Meeting: emails about meetings, schedules, departmental matters, urgent discussions

- Ignored: anything else

Return ONLY one word exactly from:

Lab_Query

Departmental_Meeting

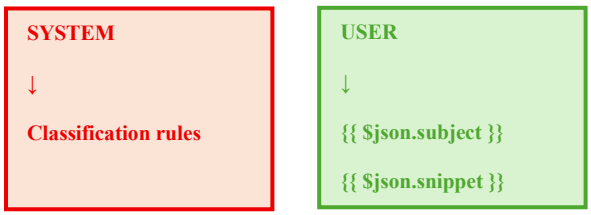
Ignored

Add user Role:

Subject: {{$json["Subject"]}}

Body:{{$json["snippet"]}}

So the structure should be:



4. Conditional Routing Logic (IF Node)

Branch A: Lab_Query (Grab text: *{{ $json["output"][0]["content"][0]["text"] }}*) → Draft

5. Response Generation Node (OpenAI Node: select "Message a Model")

Invoke an OpenAI node to generate a **professionally worded draft email** requesting missing academic identifiers and issue clarification.

System Prompt

You are an assistant of XYZ, Instructor.

Write a professional draft email asking the student to provide:

- Registration Number

- Section

- Course / Subject Code

- Clear description of the lab issue

Tone: polite, professional, clear.

persist the output as a Gmail draft only (no auto-send).

6. Create a Draft Node

Create a draft from Generated Response: {{$node["Message a model1"].json["output"][0]["content"][0]["text"].split('\n\n').slice(1).join('\n\n')}}

Set a subject: Lab Query - Missing Details

Set thread ID: {{$node["Gmail Trigger"].json["threadId"]}}

Set to email: {{$node["Gmail Trigger"].json["From"]}}

Branch B: Departmental_Meeting → Priority Label Assignment

Add a Label Node:

Set Message ID: {{$node["Gmail Trigger"].json["id"]}}

Label: High Priority

If the email pertains to departmental coordination or urgent academic matters:

- **Do NOT generate a reply**
- Apply Gmail label: **High Priority**

High Priority Definition

An email qualifies as high priority if it concerns departmental meetings, academic scheduling, urgent discussions, or requires immediate instructor attention.

Branch C: Ignored → Null Action

For all other emails:

- No draft generation
- No label assignment
- No further processing

Expected Draft Output (Lab_Query Only)

The workflow **must generate the following draft verbatim**:

Subject: Lab Query - Missing Details

Email Body:

Dear Student,

Thank you for reaching out regarding your lab query.

To assist you properly, please reply with the following details:

• Registration Number

• Section

• Course / Subject Code

• A clear description of your lab-related issue

Once these details are provided, I will be able to review your concern and guide you accordingly.

Regards,

XYZ

Instructor

Department of AI & Data Science

PART B: MODELING INTELLIGENT AGENTS & ENVIRONMENT USING PYTHON

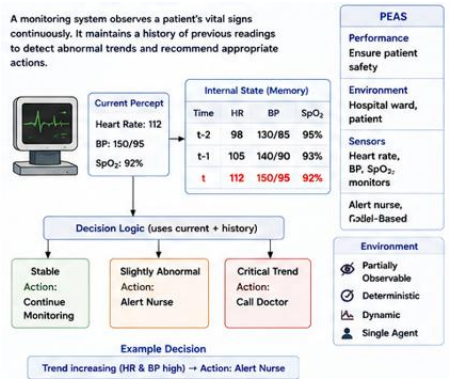
TASK 2: IMPLEMENTATION OF A MODEL-BASED AGENT FOR INTELLIGENT PATIENT MONITORING

Objective

Develop a **Model-Based Agent** capable of maintaining an internal representation of a patient's physiological state to support context-aware clinical decision making.

Problem Description

Modern hospitals employ intelligent patient monitoring systems to continuously assess physiological parameters such as heart rate, oxygen saturation, and blood pressure. Clinical decisions should not rely solely on the most recent sensor reading because transient fluctuations may not accurately represent the patient's condition. Instead, the monitoring system maintains an internal model containing previously observed measurements, enabling it to recognize persistent abnormalities and generate more reliable recommendations. Your task is to implement a Model-Based Agent that combines current observations with historical patient information to determine the appropriate clinical response.



Environment Characteristics

- Partially Observable
- Deterministic
- Dynamic

Functional Requirements

The implemented agent shall:

1. Maintain an internal memory containing previous physiological observations.
2. Continuously update the internal model after every new observation.
3. Compare current sensor readings with historical data.
4. Recommend an appropriate action (Continue Monitoring, Alert Nurse, or Call Doctor).

Expected Learning Outcome

You will learn how internal state representation enhances intelligent decision making in partially observable environments.

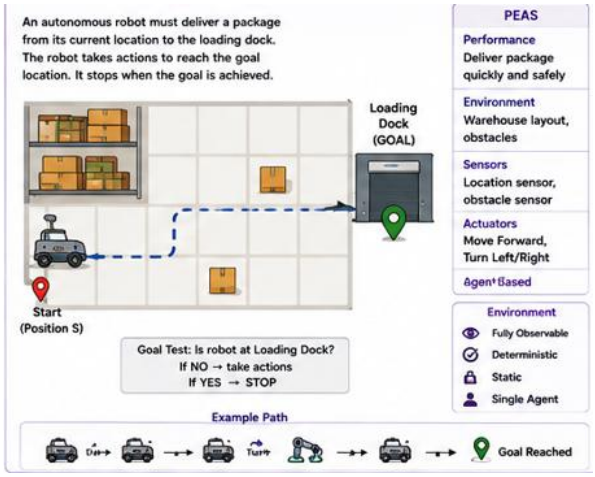
TASK 3: IMPLEMENTATION OF A GOAL-BASED AGENT FOR AUTONOMOUS WAREHOUSE LOGISTICS

Objective

Develop a **Goal-Based Agent** that selects actions according to a predefined operational objective rather than reacting solely to instantaneous environmental observations.

Problem Description

A fully automated warehouse utilizes autonomous mobile robots to transport inventory from storage racks to the loading dock for shipment. Every robot is assigned a delivery objective that must be satisfied before execution terminates. Unlike reflex-based systems, the robot continuously evaluates whether its operational goal has been achieved before selecting subsequent actions. Your task is to model the warehouse robot as a Goal-Based Agent capable of determining whether additional actions are required to accomplish the assigned delivery objective.



Environment Characteristics

- Fully Observable
- Deterministic
- Static

Functional Requirements

The implemented agent shall:

1. Define a delivery destination as the operational goal.
2. Simulate multiple initial robot locations.
3. Determine whether the current state satisfies the goal condition.
4. Continue execution until the goal state has been achieved.

Expected Learning Outcome

You will learn goal-directed reasoning by designing an intelligent agent whose behaviour is governed by explicit goal satisfaction rather than simple condition–action mappings.

UNGRADED HOME TASKS:

TASK 4: IMPLEMENTATION OF A UTILITY-BASED AGENT FOR CLOUD RESOURCE SELECTION

Objective

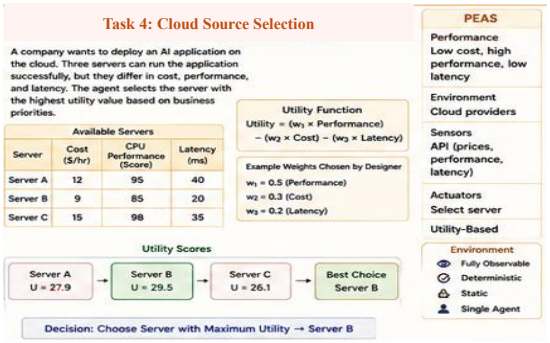
Develop a **Utility-Based Agent** that evaluates multiple feasible alternatives using a weighted utility function and recommends the optimal deployment strategy.

Problem Description

A cloud computing provider offers multiple virtual machines capable of hosting an Artificial Intelligence inference service. Although every available server satisfies the minimum

computational requirements, they differ in terms of deployment cost, computational performance, and network latency. Selecting the first available server may lead to unnecessary operational costs or degraded application performance. Therefore, an intelligent decision-making system is required to quantitatively evaluate each deployment option and recommend the most desirable alternative based on predefined business priorities. Your task is to implement a Utility-Based Agent that performs multi-criteria decision making using a designer-defined utility function.

Environment Characteristics



- Fully Observable
- Deterministic
- Static

Functional Requirements

The implemented agent shall:

1. Represent multiple candidate servers together with their attributes.
2. Define a weighted utility function reflecting organizational priorities.
3. Compute a utility score for every available deployment option.
4. Select and report the server with the highest utility value.

Expected Learning Outcome

You will learn preference-based rational decision making by integrating multiple evaluation criteria into a single quantitative utility function.

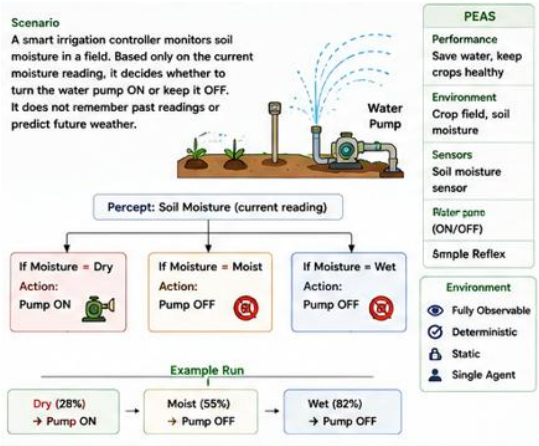
TASK 5: IMPLEMENTATION OF A SIMPLE REFLEX AGENT FOR INTELLIGENT IRRIGATION CONTROL

Objective

Design and implement a **Simple Reflex Agent** that performs instantaneous decision-making by mapping environmental percepts directly to predefined condition–action rules without maintaining any internal state or historical knowledge.

Problem Description

A precision agriculture company has deployed an automated irrigation system to optimize water consumption across multiple crop fields. The irrigation controller continuously receives soil moisture readings from distributed moisture sensors and must decide whether irrigation should be activated. Since the controller has no knowledge of previous soil conditions or future weather forecasts, every decision must be derived exclusively from the current percept. Your task is to model the irrigation controller as a **Simple Reflex Agent** and evaluate its behaviour under different environmental conditions.



Environment Characteristics

- Fully Observable
- Deterministic
- Static
- Single Agent

Functional Requirements

The implemented agent shall:

1. Receive the current soil moisture level as input.
2. Apply predefined condition–action rules to determine the appropriate response.
3. Activate or deactivate the irrigation pump accordingly.
4. Demonstrate the agent's behaviour for multiple moisture readings.

Expected Learning Outcome

You will learn the ability to design a reflex-based intelligent agent whose behaviour is solely governed by the current percept without utilizing memory, planning, or optimization mechanisms.

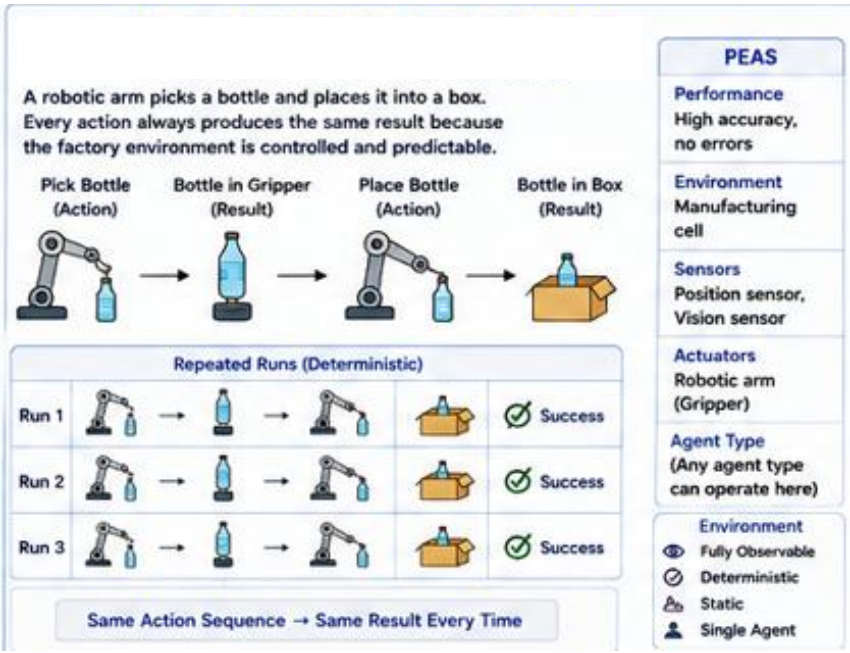
TASK 6: MODELING A DETERMINISTIC ENVIRONMENT FOR ROBOTIC MANUFACTURING

Objective

Analyze the behaviour of an intelligent agent operating within a **deterministic environment** where identical actions consistently produce identical state transitions.

Problem Description

An industrial robotic manipulator is deployed on an automated production line to perform repetitive pick-and-place operations. The manufacturing cell is carefully controlled to eliminate uncertainty, ensuring that every commanded action produces the expected physical outcome. Consequently, identical action sequences always result in identical execution trajectories. Your task is to simulate this deterministic manufacturing environment and analyze the consistency of state transitions across repeated executions.



TASK 7: MODELING A STOCHASTIC ENVIRONMENT FOR AUTONOMOUS DRONE DELIVERY

Objective

Investigate the influence of environmental uncertainty on intelligent agent behaviour within a **stochastic environment**.

Problem Description

An autonomous aerial drone is responsible for delivering medical supplies to remote healthcare facilities. The navigation controller consistently generates the same flight command for identical delivery requests; however, the operating environment is inherently uncertain due to wind disturbances, rainfall, temporary GPS degradation, and communication interference. Although the agent executes the same action, the resulting system behaviour may vary because of unpredictable environmental conditions. Your task is to model these probabilistic environmental effects and evaluate their impact on mission execution.

